

$$\begin{cases} |w+i| = \sqrt{34} \\ \arg(w-i) = \frac{3\pi}{4} \end{cases} \quad w = a+bi$$

$$a+bi-i = \frac{3\pi}{4}$$

$$(a)+(b-1)i = \frac{3\pi}{4}$$

$$\frac{b-1}{a} = \tan\left(\frac{3\pi}{4}\right) \quad \begin{matrix} -1 \\ a < 0 \\ b > 0 \end{matrix}$$

$$\frac{b-1}{a} = -1$$

$$b-1 = -a$$

$$a = 1-b \quad \Leftrightarrow \quad b = a+1$$

$$|w+i| = \sqrt{34} \quad w = a+bi$$

$$|a+bi+i| = \sqrt{34}$$

$$|(a)+(b+1)| = \sqrt{34}$$

$$a^2 + (b+1)^2 = (\sqrt{34})^2$$

$$a = 1-b$$

$$b = a+1$$

$$a^2 + (a+1+1)^2 = 34$$

$$a^2 + (a+2)^2 = 34$$

$$(a+b)^2$$

$$a^2 + a^2 + 2a \cdot 2 + 2^2 = 34$$

$$a^2 + 2ab + b^2$$

$$2a^2 + 7a + 7 = 34$$

Cuadrante

1) $0 < \theta < \frac{\pi}{2}$ $0 < a < 90$	$a > 0$	$b > 0$
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2) $\frac{\pi}{2} < \theta < \pi$ $90 < a < 180$	$a < 0$	$b > 0$
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3) $\pi < \theta < \frac{3\pi}{2}$ $180 < a < 270$	$a < 0$	$b < 0$
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4) $\frac{3\pi}{2} < \theta < 2\pi$ $270 < a < 360$	$a > 0$	$b < 0$
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En cualquier caso que el arg NO sea $\pm\pi$ u $\pm\frac{\pi}{2}$ para sacar el ángulo se $\theta = \left(\frac{b}{a} = \tan(\theta)\right)$

$$2a^2 + 7a + 7 - 37$$

$$2a^2 + 7a - 30 = 0$$

$$a = 3$$

$$a = -5$$

$$a < 0$$

$$b > 0$$

$$b = a + 1$$

$$= -5 + 1$$

$$= -4$$

NO tiene
solution

$$\begin{cases} |z| = 5 \\ \arg(3 - z) = \frac{-\pi}{2} \end{cases}$$

$$3 - (a + bi) = \frac{-\pi}{2} \quad a = 0 \quad b < 0$$

$$3 - a - bi$$

$$(3 - a) + (-b)i$$

$$3 - a = 0$$

$$-b < 0$$

$$a = 3$$

$$b > 0$$

$$|z| = 5$$

$$|a + bi| = 5$$

$$|(a) + (b)i| = 5$$

$$a^2 + b^2 = 5^2$$

$$9 + b^2 = 25 \quad b > 0$$

$$b^2 = 16$$

$$b = \pm 4 \rightarrow b = 4$$

$$|z| = 3 + 4i$$